Key to understanding supersonic radiative Marshak waves using simple models and advanced simulations

Yuval Kochavi

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1 Introduction

- Supersonic radiative Marshak waves are important in modeling energy transport in inertial confinement fusion (ICF) and astrophysical systems.
- These waves propagate faster than the material sound speed, allowing diffusion-based approximations.
- Modeling challenges include uncertainties in opacity, equations of state, radiation losses, and multidimensional effects.
- This study introduces:
  - A semi-analytic model based on diffusion theory which takes the following into account:
    - 1D radiative diffusion,
    - Radiation losses to gold walls,
    - Ablated wall material and motion,
    - Diagnostic-based effective front correction.
  - Detailed two-dimensional implicit Monte Carlo (IMC) simulations,
  - A unified framework applicable across diverse experimental setups.

2 Experimental Overview

- Experiments use laser-driven gold hohlraums to generate X-rays, which drive radiative waves into attached low-density foam tubes.
- Key parameters:
  - Laser energy: 1–350 kJ
  - Radiation temperature(T_D): 85–310 eV

- Foam densities: 10–160 mg/cm^3

- Two primary diagnostic strategies are used to determine the heat front position:
  - **Variable foam length method:** The foam length is changed between shots. The breakout time is recorded for each length, and the heat front velocity is inferred. The breakout is typically identified as the first sign of signal (e.g., darkening) at the rear end of the foam.
  - **Slit imaging method:** A single foam-filled gold cylinder includes either multiple short slits or one long slit along its side. The heat front's position is observed directly over time through the slit(s), allowing single-shot tracking. The main challenge is achieving sufficient time resolution to determine when the front passes each slit location.

- Despite varying geometries and techniques, all experiments probe radiation-dominated, supersonic heat transport.

3 Theoretical Background

- Governing equation: radiation diffusion in LTE,

$$\frac{\partial U}{\partial t} = \frac{\partial}{\partial x}\left(\frac{c}{3\rho\kappa_R}\frac{\partial U}{\partial x}\right)$$

- Opacity and heat capacity follow power laws:

$$\kappa \propto T^{-\alpha}, \quad C_v \propto T^{\beta}$$

- Semi-analytic model accounts for:
  - 1D radiative diffusion,
  - Radiation losses to gold walls,
  - Ablated wall material and motion,
  - Diagnostic-based effective front correction.

4 Simple Semi-Analytic Model

4.1 1D Diffusion Model (Hammer–Rosen Formulation)

The foundational model for the radiative Marshak wave is based on the Hammer–Rosen (HR) solution from 2003.

- Assumes optically thick foam and local thermodynamic equilibrium (LTE).
- Reduces radiation transport to a nonlinear diffusion equation for $T(x,t)$.

- Opacity and internal energy modeled as power laws:

$$\kappa^{-1} = gT^a\rho^{-\lambda}, \quad e = fT^b\rho^{-\mu}$$

- Self-similar solution from Hammer–Rosen yields front position:

$$x_F^2(t) = \frac{2+\varepsilon}{1-\varepsilon} CH^{-\varepsilon}(t)\int_0^t H(t')dt' \tag{1}$$

with:

$$H = T_s^{4+\alpha}, \varepsilon = \frac{\beta}{4+\alpha}, C = \frac{16g\sigma}{3f(4+\alpha)\rho^{2-\mu-\lambda}}, \tag{2}$$

- Temperature profile behind the front approximated by a Henyey-like expression (it is pretty good approximation when the material is Low Z and so ε is small):

$$T(x,t) = T_S(t)(1-\frac{x}{x_F})^{\frac{1}{4+\alpha-\beta}} \tag{3}$$

4.2 Surface vs. Drive Temperature: Boundary Condition Correction

- The drive temperature T_D (from the hohlraum) is not equal to the foam surface temperature T_s.
- Marshak boundary condition relates incoming flux and re-emission:

$$\sigma_{\mathrm{SB}} T_D^4(t) = \sigma_{\mathrm{SB}} T_S^4(t) + \frac{F(0,t)}{2},$$

where $F(0,t)$ is the radiation flux into the foam at $x = 0$, and also the derivative of the energy stored in the foam:

$$F(0,t) = \frac{d}{dt}\int_0^{x_f(t)} \rho e(x,t)dx = \frac{d}{dt}\left[f\rho^{1-\mu}x_F(t)H^{\varepsilon}(t)(1-\varepsilon)\right]$$

- A system of equations (HR front, energy stored in foam, Marshak condition) is solved numerically for $T_s(t), x_F(t)$. Detailed explanation in the $aricle1.tex$ sebsection 5.2.
- This correction reduces T_s and slows $x_f(t)$, improving agreement with experiments.

4.3 Diagnostic Cutoff Correction

- Experiments define the front as where flux reaches a fraction χ of its peak.
- Apply a correction using the modeled temperature profile:

$$T(x,t) = T_s(t)\left(1-\frac{x}{x_f(t)}\right)^{\frac{1}{4+\alpha-\beta}}$$

- Solving $T(x'_f) = T_s f^{1/4}$ gives:

$$x'_f(t) = x_f(t)\left(1 - f^{(4+\alpha-\beta)/4}\right)$$

- This reduces the effective front position, which we get from the simulation. And using this connection, one get the real front position, which is the one that should be compared to the experiments.

4.4 2D Effects: Wall Energy Loss and Ablation

Wall Energy Loss

- In cylindrical experiments, the low-Z foam is enclosed within a high-Z wall (typically gold), which absorbs a portion of the radiation propagating through the foam.

- As the Marshak wave progresses axially through the foam, radiation leaks laterally into the surrounding wall, driving a 1D subsonic Marshak wave into the wall material.

- To model this, each axial slice of the foam is treated as emitting radiation sideways into a semi-infinite wall. The wall absorbs this flux and stores energy based on the time it has been exposed. Each spatial element is exposed to the radiation only after the heat front reaches it at time $t_0(x)$ for a period of $t - t_0(x)$.

- The total energy lost to the wall is calculated by integrating over all axial slices exposed to the heat front. the expression is:

$$E_{\text{wall}}(t) = 2\pi R \int_0^{x_F(t)} \int_{t_0(x)}^{t} \dot{E}_{material}(t')dt'dx$$

- The model uses analytic expressions from Shussman and Heizler (2015) to evaluate energy deposition into gold walls with LTE radiative diffusion assumptions.

- For experiments with different wall materials:

  – For **gold walls**, constants are taken from CRSTA opacity/EOS data.

  $$E_{gold}(t) = 0.59T_0^{3.35}(t - t_0(x))^{0.59}[hJ/mm^2]$$

  – For **beryllium walls**, which are more transparent and supersonic-like, alternate constants are used.

  $$E_{Be}(t) = 1.27T_0^{4.99}(t - t_0(x))^{0.5}[hJ/mm^2]$$

  – For **bare foam** (no wall), lateral radiation is assumed to freely stream into vacuum (i.e., Marshak vacuum boundary condition).

- The total energy loss to the wall, $E_{\text{wall}}(t)$, reduces the energy remaining in the foam, thereby decreasing the surface temperature $T_s(t)$ (in about $10eV$).
- This effect causes a further slowdown of the Marshak wavefront. For example, in the Back et al. experiment, wall losses accounted for most of the gap between the 1D prediction and the observed wavefront position.

Wall Ablation and Foam Compression

- As the foam's surface is heated by the radiation wave, the inner surface of the gold wall is also exposed to intense radiation. This drives a subsonic Marshak wave into the gold, heating it and causing the gold to ablate inward into the foam channel.
- The wall ablation speed $u_{\text{abl}}(t)$ is calculated using the subsonic Marshak wave solution into a semi-infinite medium, derived by Shussman and Heizler (2015). For gold walls, the ablation speed in vacuum is given by:

  $$u_{\text{abl}}(t) = -510.1\tilde{u}(\rho_{gold})T_0^{0.716}(t - t_0(x))^{0.036}[km/s] \tag{4}$$

  The ablation front inside the gold wall is not uniform; it follows a self-similar velocity profile $\tilde{u}(\rho)$ ranging from 1 at the surface to 0 deeper inside the wall.
- Because the gold wall is restrained by the foam, the model selects $\tilde{u}$ at a density corresponding to approximately $\rho \approx 4\rho_{\text{foam}}$, based on 2D simulation calibration.
- In the case of low-Z walls such as beryllium sleeves, where the internal heat wave is nearly supersonic, the model sets $u_{\text{abl}} = 0$, effectively ignoring wall ablation in those cases.
- The foam channel radius decreases over time due to wall ablation:

  $$R(t, x) = R_0 - u_{\text{abl}}(t)(t - t_0(x))$$

  which effectively reduces the cross-sectional area available for energy delivery.
- As a result, the total incoming energy power is reduced:

  $$\begin{aligned} E_{\text{in}} &= \int_0^t F(t', 0) \cdot \pi R^2(t', x)\, dt' \\ &= \int_0^t 2\sigma_{\text{SB}} \left(T_D^4(t') - T_s^4(t')\right) \cdot \pi R^2(t', x)\, dt' \\ &= 2\pi\sigma_{\text{SB}} \int_0^t \left(T_D^4(t') - T_s^4(t')\right) \cdot \left(R_0^2 - u_{\text{abl}}(t')t'\right)\, dt' \end{aligned} \tag{5}$$

  This leads to a further reduction in $T_s(t)$, since less energy reaches the foam.

- Simultaneously, ablated wall mass enters and mixes with the foam, increasing its effective average density over time. The evolving foam density is approximated as:

$$\rho_{\text{eff}}(t) = \rho_0 \frac{V_0}{V(t)} = \frac{\rho_0 V_0}{\int_0^{x_f(t)} \pi R^2(t,x)dx} \tag{6}$$

  Watch that $C(t)$ changes because it depends on ρ:

$$C(t) = \frac{16g\sigma}{3f(4+\alpha)t} \int_0^t \rho_{\text{eff}}^{\mu+\lambda-2}(t')dt' \tag{7}$$

- Since radiative transport is sensitive to density (through opacity and heat capacity), the increasing ρ_{eff} further slows the wavefront.
- The model updates the Hammer–Rosen front position formula by incorporating both the time-dependent drive area and the evolving foam density, brings the semi-analytic prediction into near-exact agreement with measured data.

5 Comparison with Experiments

Massen et al. (1994, GEKKO-XII)

- Target: Lead-doped plastic foam ($C_{11}H_{16}Pb_{0.3852}$), density $\rho \approx 80$ mg/cm^3.
- Drive: Hohlraum-driven radiation at $T_D \approx 100$, 120, and 150 eV using 0.8–0.9 ns laser pulses.
- Foam lengths: 50–300 µm; no explicit gold wall noted.
- Diagnostics: X-ray streak camera measured breakout time at foam rear.
- Front definition: Breakout timing, no flux cutoff reported.
- Modeling: 1D semi-analytic model with corrected $T_s(t)$ (Marshak boundary) accurately reproduced front trajectories.
- Notes: Classical HR model overestimated velocity by $\sim$20%; corrected model explains this without tuning.

Back et al. (1995, OMEGA Low-Energy)

- Target: SiO_2 foam, density $\rho \approx 10$ mg/cm^3.
- Drive: Radiation temperature $T_D \approx 85$ eV with 10 ns pulse.
- Foam lengths: 0.5–1.5 mm; cylindrical gold wall.
- Diagnostics: Emergent flux vs. time measured for breakout.

- Front definition: Likely flux onset, no cutoff specified.
- Modeling: With updated opacity/EOS for low-T and low-ρ, 2D semi-analytic and IMC models matched breakout timing.
- Notes: Energy regime emphasizes opacity modeling; flux shape agreement was qualitative.

Back et al. (2000, OMEGA High-Energy)

- Target: SiO_2 and Ta_2O_5 foams, densities $\rho \approx 50$ and 40 mg/cm^3, respectively.
- Drive: $T_D \approx 190$ eV with up to 1 ns laser energy.
- Foam lengths: 0.25–1.25 mm; 1.6 mm-diameter cylindrical gold hohlraum.
- Diagnostics: X-ray flux breakout timing vs. length.
- Front definition: Point where flux reached 50% of maximum.
- Modeling: Required full 2D treatment—wall loss and ablation—to reproduce measured front speed and flux history.
- Be wall case: Beryllium sleeve produced ∼10% slower wave; matched by modified model.

Keiter et al. (2007–2013, OMEGA)

- Target: $C_{15}H_{20}O_6$ foams, $\rho \approx 62$–65 mg/cm^3, with and without 12% Au nanoparticle doping.
- Drive: $T_D \approx 200$–210 eV.
- Geometry: 1.2 mm long, 0.8 mm diameter gold-coated tubes with side window.
- Diagnostics: Side-on X-ray emission through gold slit.
- Front definition: Position at 40% of peak radiative flux.
- Modeling: Required front cutoff correction + 2D energy loss + wall ablation to match data.
- Notes: Doped foams showed slower front due to increased opacity; model reproduced both doped and pure cases.

Ji-Yan et al. (2010, SG-II)

- Target: Polystyrene foam (C_8H_8), $\rho \approx 160$ mg/cm^3, no confining wall.
- Drive: $T_D \approx 175$ eV.
- Geometry: 0.3 mm long, 0.2 mm diameter.
- Diagnostics: Side-on self-emission through vacuum.
- Front definition: Flux reaching 50% of peak value.
- Modeling: Used vacuum lateral boundary condition to account for radiative escape.
- Notes: 2D lateral energy loss essential even without walls; model and IMC both agreed with experiment.

Moore et al. (2015, NIF "Pleiades")

- Targets: SiO_2 foam ($\rho \approx 100$ mg/cm^3) and Cl-doped CH (C_8H_7Cl, $\rho \approx 100$ mg/cm^3).
- Drive: $T_D \approx 300$ eV with a foam length of 2.8 mm.
- Geometry: 2 mm diameter gold tube with side windows.
- Diagnostics:
  - Breakout timing via Dante x-ray diodes,
  - Front tracking with side-on framing camera.
- Front definition: Position vs. time from camera data.
- Modeling:
  - HR model overpredicted front speed.
  - Full 2D model (Marshak T_s, wall loss, ablation, cutoff correction) reproduced trajectories within 5–10%.
- Notes: IMC simulations matched both centerline and edge front speeds; NIF scale required all corrections.

Overall Summary

- All experiments spanning 1994–2015, with densities of 10–160 mg/cm^3 and T_D from 85–300 eV, were matched by a single unified semi-analytic model.
- 1D models were adequate only in short, low-energy cases.

- 2D effects—especially wall loss, foam channel narrowing, and diagnostic cutoff correction—were essential for matching timing and trajectory in all long or high-flux experiments.